Trigger_Part_2

HARSH KOTHADIYA

AFTER DML TRIGGER

Part:1

```sql
--PART:1 DATABASE CREATE
CREATE DATABASE Trigger_DB;

USE Trigger_DB
```

Part:2

```sql
--PART:2 TABLE CREATE
CREATE TABLE CRICKETER_DETAIL(
    CDID INT IDENTITY(1,1) PRIMARY KEY,
    FIRST_NAME VARCHAR(20),
    LAST_NAME VARCHAR(20),
    DOB DATE,
    ROLE VARCHAR(50),
    HIGHEST_T20I_RUNS INT,
    CREATEDAT DATETIME DEFAULT GETDATE()
);
```

Part:3

```sql
--PART:3 TWO ROW DATA INSERT
INSERT INTO CRICKETER_DETAIL(FIRST_NAME,LAST_NAME,DOB,ROLE,HIGHEST_T20I_RUNS)
VALUES('ROHIT','SHARMA','1987-04-30','Batter',121)

INSERT INTO CRICKETER_DETAIL(FIRST_NAME,LAST_NAME,DOB,ROLE,HIGHEST_T20I_RUNS)
VALUES('VIRAT','KOHLI','1988-11-05','Batter',122)
```

98% 12 0

Messages

```
(1 row affected)

(1 row affected)

Completion time: 2026-04-26T10:11:29.5250113+05:30
```

Part:4

```sql
--PART:4 DATA GET
SELECT * FROM CRICKETER_DETAIL;
```

98 % 1 0

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	121	2026-04-26 10:11:29.503
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513

Part:5

```sql
--PART:5
--UPDATEDAT COLUMN MISSING TABLE CREATE TIME SO NOW ADD THIS COLUMN

ALTER TABLE CRICKETER_DETAIL
ADD UPDATEDAT DATETIME;
```

```sql
SELECT * FROM CRICKETER_DETAIL;
```

98 % 1 0

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	121	2026-04-26 10:11:29.503	NULL
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL

Part:6

```sql
--PART:6 NOW CDID = 1 --> HIGHEST_T20I_RUNS UPDATE 150
UPDATE CRICKETER_DETAIL
SET HIGHEST_T20I_RUNS = 150
WHERE CDID = 1;
```

Part:7

BEFORE

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	121	2026-04-26 10:11:29.503	NULL
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL

AFTER

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	150	2026-04-26 10:11:29.503	NULL
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL

```
/*
ONLY HIGHEST_T20I_RUNS UPDATE WHEN UPDATE QUERY RUN
BUT UPDATEDAT COLUMN STILL NULL
SO NOW MY REQUIREMENT WHENEVER UPDATE  QUERY RUN UPDATEDAT
COLUMN UPDATE AUTO SO USE TRIGGER
*/
```

Part:8

```sql
--PART:8
CREATE TRIGGER TRG_CRICKETER_DETAIL_UPDATEDAT
ON CRICKETER_DETAIL
AFTER UPDATE
AS
BEGIN
    SET NOCOUNT ON;

    UPDATE C
    SET UPDATEDAT = GETDATE()
    FROM CRICKETER_DETAIL C
    INNER JOIN INSERTED I
        ON C.CDID = I.CDID;

    PRINT 'TRIGGER FIRED: TRG_CRICKETER_DETAIL_UPDATEDAT.'
END;
```

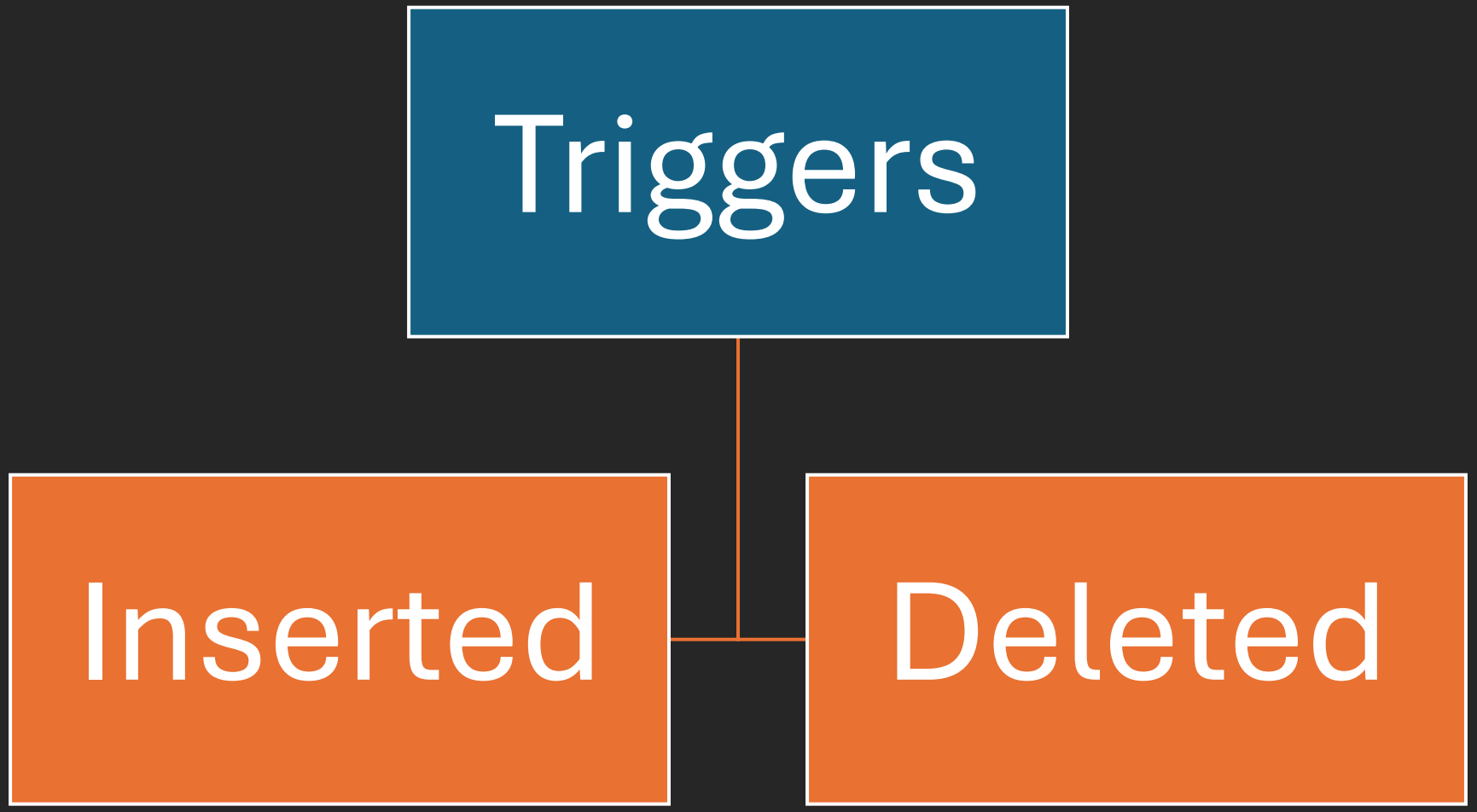
Part:9

```sql
SELECT 'BEFORE UPDATE 200',* FROM CRICKETER_DETAIL;

--PART:9 NOW CDID = 1 --> HIGHEST_T20I_RUNS UPDATE 200
UPDATE CRICKETER_DETAIL
SET HIGHEST_T20I_RUNS = 200
WHERE CDID = 1;

SELECT 'AFTER UPDATE 200',* FROM CRICKETER_DETAIL;
```

	(No column name)	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	BEFORE UPDATE 200	1	ROHIT	SHARMA	1987-04-30	Batter	150	2026-04-26 10:11:29.503	NULL
2	BEFORE UPDATE 200	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL

	(No column name)	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	AFTER UPDATE 200	1	ROHIT	SHARMA	1987-04-30	Batter	200	2026-04-26 10:11:29.503	2026-04-26 10:31:32.940
2	AFTER UPDATE 200	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL

Part:10



🧠 Simple understanding:

Operation	Inserted Table	Deleted Table
INSERT	New data	Empty
DELETE	Empty	Old data
UPDATE	New values	Old values

📌 One-line summary:

👉 *Inserted = New Data*

👉 *Deleted = Old Data*

```
/*
Inserted and Deleted tables in SQL Server are temporary, system-generated tables used
inside triggers.
*/
```

Part:11

```sql
--PART:11 INSERT TRIGGER FOR Inserted and Deleted Tables
CREATE TRIGGER TRG_INSERT_ON_CRICKETER_DETAIL
ON CRICKETER_DETAIL
AFTER INSERT
AS
BEGIN
    SET NOCOUNT ON;

    SELECT 'INSERTED',* FROM INSERTED;
    SELECT 'DELETED',* FROM DELETED;

    PRINT 'TRIGGER FIRED: TRG_INSERT_ON_CRICKETER_DETAIL.'

END;
--GO
```

```sql
--NOW INSERT DATA
INSERT INTO CRICKETER_DETAIL
(FIRST_NAME, LAST_NAME, DOB,ROLE, HIGHEST_T20I_RUNS)
VALUES
('Yuvraj','Singh','1981-12-12', 'All-Rounder',77),
('Gautam','Gambhir','1981-10-14', 'Batsman',75),
('Virender','Sehwag','1978-10-20', 'Batsman',68),
('Ajinkya','Rahane','1988-06-06', 'Batsman',61),
('Shikhar','Dhawan','1985-12-05', 'Batsman',92),
('KL','Rahul','1992-04-18', 'Wicketkeeper-Batsman',110);
```

Results | Messages

	(No column name)	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	INSERTED	32	KL	Rahul	1992-04-18	Wicketkeeper-Batsman	110	2026-04-26 11:17:57.010	NULL
2	INSERTED	31	Shikhar	Dhawan	1985-12-05	Batsman	92	2026-04-26 11:17:57.010	NULL
3	INSERTED	30	Ajinkya	Rahane	1988-06-06	Batsman	61	2026-04-26 11:17:57.010	NULL
4	INSERTED	29	Virender	Sehwag	1978-10-20	Batsman	68	2026-04-26 11:17:57.010	NULL
5	INSERTED	28	Gautam	Gambhir	1981-10-14	Batsman	75	2026-04-26 11:17:57.010	NULL
6	INSERTED	27	Yuvraj	Singh	1981-12-12	All-Rounder	77	2026-04-26 11:17:57.010	NULL

(No column name)	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT

Results | Messages

```
TRIGGER FIRED: TRG_INSERT_ON_CRICKETER_DETAIL.

(6 rows affected)

Completion time: 2026-04-26T11:17:57.1202024+05:30
```

```
2  SELECT * FROM CRICKETER_DETAIL
```

192 % 1 0 Ln: 2, Ch: 1 TABS

Results Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	200	2026-04-26 10:11:29.503	2026-04-26 10:31:32.940
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL
3	27	Yuvraj	Singh	1981-12-12	All-Rounder	77	2026-04-26 11:17:57.010	NULL
4	28	Gautam	Gambhir	1981-10-14	Batsman	75	2026-04-26 11:17:57.010	NULL
5	29	Virender	Sehwag	1978-10-20	Batsman	68	2026-04-26 11:17:57.010	NULL
6	30	Ajinkya	Rahane	1988-06-06	Batsman	61	2026-04-26 11:17:57.010	NULL
7	31	Shikhar	Dhawan	1985-12-05	Batsman	92	2026-04-26 11:17:57.010	NULL
8	32	KL	Rahul	1992-04-18	Wicketkeeper-Batsman	110	2026-04-26 11:17:57.010	NULL

Part:12

```sql
--PART:12 UPDATE TRIGGER FOR Inserted and Deleted Tables
CREATE TRIGGER TRG_UPDATE_ON_CRICKETER_DETAIL
ON CRICKETER_DETAIL
AFTER UPDATE
AS
BEGIN
    SET NOCOUNT ON;

    SELECT 'INSERTED',* FROM INSERTED;
    SELECT 'DELETED',* FROM DELETED;

    PRINT 'TRIGGER FIRED: TRG_UPDATE_ON_CRICKETER_DETAIL.'

END;
--GO
```

```sql
--NOW UPDATE DATA
UPDATE CRICKETER_DETAIL
SET HIGHEST_T20I_RUNS = 152
WHERE FIRST_NAME = 'KL' AND LAST_NAME = 'Rahul';
```

```
TRIGGER FIRED: TRG_UPDATE_ON_CRICKETER_DETAIL.
TRIGGER FIRED: TRG_CRICKETER_DETAIL_UPDATEDAT.
TRIGGER FIRED: TRG_UPDATE_ON_CRICKETER_DETAIL.
```

Results | Messages

	(No column name)	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	INSERTED	32	KL	Rahul	1992-04-18	Wicketkeeper-Batsman	152	2026-04-26 11:17:57.010	2026-04-26 11:19:09.1

TRG_CRICKETER_DETAIL_UPDATEDAT

	(No column name)	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	DELETED	32	KL	Rahul	1992-04-18	Wicketkeeper-Batsman	152	2026-04-26 11:17:57.010	NULL

	(No column name)	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	INSERTED	32	KL	Rahul	1992-04-18	Wicketkeeper-Batsman	152	2026-04-26 11:17:57.010	NULL

	(No column name)	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	DELETED	32	KL	Rahul	1992-04-18	Wicketkeeper-Batsman	110	2026-04-26 11:17:57.010	NULL

TRG_UPDATE_ON_CRICKETER_DETAIL

Part:13

```sql
--PART:13 DELETE TRIGGER FOR Inserted and Deleted Tables
CREATE TRIGGER TRG_DELETE_ON_CRICKETER_DETAIL
ON CRICKETER_DETAIL
AFTER DELETE
AS
BEGIN
    SET NOCOUNT ON;

    SELECT 'INSERTED',* FROM INSERTED;
    SELECT 'DELETED',* FROM DELETED;

    PRINT 'TRIGGER FIRED: TRG_DELETE_ON_CRICKETER_DETAIL.'

END;
--GO
```

```sql
--NOW DELETE DATA
DELETE FROM CRICKETER_DETAIL
WHERE CDID NOT IN (1,2);
```

9% No issues found Ln: 5, Ch: 25 TABS CRLF UTF-8

Results Messages

	(No column name)	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
	DELETED	32	KL	Rahul	1992-04-18	Wicketkeeper-Batsman	152	2026-04-26 11:17:57.010	2026-04-26 11:19:0
2	DELETED	31	Shikhar	Dhawan	1985-12-05	Batsman	92	2026-04-26 11:17:57.010	NULL
3	DELETED	30	Ajinkya	Rahane	1988-06-06	Batsman	61	2026-04-26 11:17:57.010	NULL
4	DELETED	29	Virender	Sehwag	1978-10-20	Batsman	68	2026-04-26 11:17:57.010	NULL
5	DELETED	28	Gautam	Gambhir	1981-10-14	Batsman	75	2026-04-26 11:17:57.010	NULL
6	DELETED	27	Yuvraj	Singh	1981-12-12	All-Rounder	77	2026-04-26 11:17:57.010	NULL

```
2  SELECT * FROM CRICKETER_DETAIL
3
```

19% 1 0 Ln: 2

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	200	2026-04-26 10:11:29.503	2026-04-26 10:31:32.940
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL

Part:14

```sql
--PART:14 NOW ABOVE INSERT,UPDATE,DELETE TRIGGER DROP
DROP TRIGGER TRG_INSERT_ON_CRICKETER_DETAIL;
DROP TRIGGER TRG_UPDATE_ON_CRICKETER_DETAIL;
DROP TRIGGER TRG_DELETE_ON_CRICKETER_DETAIL;
```

98 % No issues found

Messages

```
Commands completed successfully.

Completion time: 2026-04-26T11:29:49.2285919+05:30
```

Part:15

```sql
--PART:15 ONLY HIGHEST_T20I_RUNS COLUMN INSERT VALUE SO
  (TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_INSERT) TRIGGER FIRED
/*
    THIS TRIGGER DATA STORE TABLE: HIGHEST_T20I_RUNS_INSERT_LOG
*/
CREATE TABLE HIGHEST_T20I_RUNS_INSERT_LOG(
    ID INT IDENTITY(1,1) PRIMARY KEY,
    CDID INT,
    HIGHEST_T20I_RUNS INT,
    CREATEDAT DATETIME DEFAULT GETDATE()
)
```

```sql
CREATE TRIGGER TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_INSERT
ON CRICKETER_DETAIL
AFTER INSERT
AS
BEGIN
    IF EXISTS(
        SELECT 1
        FROM INSERTED
        WHERE HIGHEST_T20I_RUNS IS NOT NULL
    )
    BEGIN
        PRINT 'TRIGGER FIRED: TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_INSERT.'

        INSERT INTO HIGHEST_T20I_RUNS_INSERT_LOG(CDID,HIGHEST_T20I_RUNS)
        SELECT CDID,HIGHEST_T20I_RUNS FROM INSERTED
    END
END
```

```sql
--BELOW ROW INSERT --> TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_INSERT --> FIRED
INSERT INTO CRICKETER_DETAIL(FIRST_NAME,LAST_NAME,DOB,ROLE,HIGHEST_T20I_RUNS)
VALUES('Yashasvi','Jaiswal','2001-12-28','Opening Batter',100)
```

Messages

```
TRIGGER FIRED: TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_INSERT.

(1 row affected)
```

```
SELECT * FROM CRICKETER_DETAIL
```

119% 1 0 Ln: 3, Ch

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	200	2026-04-26 10:11:29.503	2026-04-26 10:31:32.940
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL
3	33	Yashasvi	Jaiswal	2001-12-28	Opening Batter	100	2026-04-26 12:10:49.323	NULL

```
SELECT * FROM HIGHEST_T20I_RUNS_INSERT_LOG;
```

119% 1 0

Results | Messages

	ID	CDID	HIGHEST_T20I_RUNS	CREATEDAT
1	1	33	100	2026-04-26 12:10:49.337

```sql
--BELOW ROW INSERT --> TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_INSERT --> NOT FIRED
INSERT INTO CRICKETER_DETAIL(FIRST_NAME,LAST_NAME,DOB,ROLE)
VALUES('Rinku','Singh','1997-10-12','Middle-order Batter')
```

Messages

```
(1 row affected)
```

```
SELECT * FROM CRICKETER_DETAIL
```

19% 1 0 Ln: 3, Ch: 1

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	200	2026-04-26 10:11:29.503	2026-04-26 10:31:32.940
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL
3	33	Yashasvi	Jaiswal	2001-12-28	Opening Batter	100	2026-04-26 12:10:49.323	NULL
4	34	Rinku	Singh	1997-10-12	Middle-order Batter	NULL	2026-04-26 12:11:57.750	NULL

```
SELECT * FROM HIGHEST_T20I_RUNS_INSERT_LOG;
```

this case HIGHEST_T20I_RUNS not insert so trigger not fired so not insert data in HIGHEST_T20I_RUNS_INSERT_LOG this table

119% 1 0

Results | Messages

	ID	CDID	HIGHEST_T20I_RUNS	CREATEDAT
1	1	33	100	2026-04-26 12:10:49.337

Part:16

```
--PART:16 ONLY HIGHEST_T20I_RUNS COLUMN UPDATE VALUE SO
(TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_UPDATE) TRIGGER FIRED
/*
    THIS TRIGGER DATA STORE TABLE: HIGHEST_T20I_RUNS_UPDATE_LOG
    VALUE CHANGE SO TRIGGER FIRED
    SAME VALUE UPDATE SO NOT FIRED -- MY REQUIREMENT
*/
```

```sql
CREATE TABLE HIGHEST_T20I_RUNS_UPDATE_LOG(
    ID INT IDENTITY(1,1) PRIMARY KEY,
    CDID INT,
    OLD_HIGHEST_T20I_RUNS INT,
    NEW_HIGHEST_T20I_RUNS INT,
    CREATEDAT DATETIME DEFAULT GETDATE()
)
```

```sql
CREATE TRIGGER TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_UPDATE
ON CRICKETER_DETAIL
AFTER UPDATE
AS
BEGIN
    IF UPDATE(HIGHEST_T20I_RUNS) --HIGHEST_T20I_RUNS ONLY THIS COLUMN VALUE CHANGE SO THIS TRIGGER FIRED
    BEGIN
        IF EXISTS(
            SELECT 1
            FROM INSERTED AS I
            INNER JOIN DELETED AS D
            ON I.CDID = D.CDID
            WHERE   (I.HIGHEST_T20I_RUNS <> D.HIGHEST_T20I_RUNS)
                OR (I.HIGHEST_T20I_RUNS IS NULL AND D.HIGHEST_T20I_RUNS IS NOT NULL)
                OR (I.HIGHEST_T20I_RUNS IS NOT NULL AND D.HIGHEST_T20I_RUNS IS NULL)
        )
        BEGIN
            PRINT 'TRIGGER FIRED: TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_UPDATE.'

            INSERT INTO HIGHEST_T20I_RUNS_UPDATE_LOG(CDID,OLD_HIGHEST_T20I_RUNS,NEW_HIGHEST_T20I_RUNS)
            SELECT I.CDID,D.HIGHEST_T20I_RUNS,I.HIGHEST_T20I_RUNS
            FROM INSERTED AS I
            INNER JOIN DELETED AS D
            ON I.CDID = D.CDID
            WHERE    (I.HIGHEST_T20I_RUNS <> D.HIGHEST_T20I_RUNS)
                OR (I.HIGHEST_T20I_RUNS IS NULL AND D.HIGHEST_T20I_RUNS IS NOT NULL)
                OR (I.HIGHEST_T20I_RUNS IS NOT NULL AND D.HIGHEST_T20I_RUNS IS NULL)
        END
    END
END
```

```sql
--BELOW ROW UPDATE --> TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_UPDATE --> FIRED
UPDATE CRICKETER_DETAIL
SET HIGHEST_T20I_RUNS = 121
WHERE CDID = 1
```

```sql
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 1 --CDID = 1 HIGHEST_T20I_RUNS VALUE  CHANGE
SELECT * FROM HIGHEST_T20I_RUNS_UPDATE_LOG WHERE CDID = 1 --CDID = 1  LOG  INSERT
```

119% 2 0 Ln: 3, Ch: 58 TABS CRLF U

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	121	2026-04-26 10:11:29.503	2026-04-26 12:21:05.463

	ID	CDID	OLD_HIGHEST_T20I_RUNS	NEW_HIGHEST_T20I_RUNS	CREATEDAT
1	1	1	200	121	2026-04-26 12:21:05.477

```sql
--BELOW ROW UPDATE --> TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_UPDATE --> NOT FIRED --> BECAUSE FIRST_NAME Update
UPDATE CRICKETER_DETAIL
SET FIRST_NAME = 'Yashasvi Jaiswal'
WHERE CDID = 33
```

```sql
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 33  --CDID = 33 FIRST_NAME VALUE CHANGE
SELECT * FROM HIGHEST_T20I_RUNS_UPDATE_LOG WHERE CDID = 33--CDID = 33 NOT  LOG INSERT
```

108 % 2 0 Ln: 2, Ch: 86

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	33	Yashasvi Jaiswal	Jaiswal	2001-12-28	Opening Batter	100	2026-04-26 12:10:49.323	2026-04-26 12:26:44.230

ID	CDID	OLD_HIGHEST_T20I_RUNS	NEW_HIGHEST_T20I_RUNS	CREATEDAT

```
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 2
```

108 % 2 0

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	NULL

```
--BELOW ROW UPDATE --> TRG_CRICKETER_DETAIL_HIGHEST_T20I_RUNS_UPDATE --> NOT FIRED -->BECAUSE SAME HIGHEST_T20I_RUNS
UPDATE CRICKETER_DETAIL
SET HIGHEST_T20I_RUNS = 122
WHERE CDID = 2
```

```sql
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 2 --CDID = 2 HIGHEST_T20I_RUNS VALUE NOT  CHANGE
SELECT * FROM HIGHEST_T20I_RUNS_UPDATE_LOG WHERE CDID = 2 --CDID = 2  LOG NOT INSERT
```

108 % 4 0 Ln: 3, Ch: 1 TABS CRLF

Results Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	2026-04-26 12:30:28.823

ID	CDID	OLD_HIGHEST_T20I_RUNS	NEW_HIGHEST_T20I_RUNS	CREATEDAT

Part:17

```sql
--PART:17 INSERT DATA
INSERT INTO CRICKETER_DETAIL(FIRST_NAME,LAST_NAME,DOB,ROLE,HIGHEST_T20I_RUNS)
VALUES('Axar','Patel','1994-01-20','All-rounder',65)
```

```sql
1  SELECT * FROM CRICKETER_DETAIL
```

108% 1 0 Ln: 1, Ch: 31 TAB:

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	Rohit	SHARMA	1987-04-30	Batter	121	2026-04-26 10:11:29.503	2026-04-26 12:24:37.610
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	2026-04-26 12:30:28.823
3	33	Yashasvi Jaiswal	Jaiswal	2001-12-28	Opening Batter	100	2026-04-26 12:10:49.323	2026-04-26 12:26:44.230
4	34	Rinku	Singh	1997-10-12	Middle-order Batter	NULL	2026-04-26 12:11:57.750	NULL
5	35	Axar	Patel	1994-01-20	All-rounder	65	2026-04-26 12:32:00.853	NULL

Part:18

```sql
--PART:18
--NOW DELETE TRIGGER
/*
    THIS TRIGGER DATA STORE TABLE:  CRICKETER_DETAIL ROW DELETE SO DATA ENTER DELETE_LOG TABLE
*/
```

```sql
CREATE TABLE DELETE_LOG(
    ID INT IDENTITY(1,1) PRIMARY KEY,
    COLUMN_LIST NVARCHAR(MAX),
    CREATEDAT DATETIME DEFAULT GETDATE()
)
```

```sql
CREATE TRIGGER TRG_CRICKETER_DETAIL_DELETE
ON CRICKETER_DETAIL
AFTER DELETE
AS
BEGIN
    PRINT 'TRIGGER FIRED: TRG_CRICKETER_DETAIL_DELETE.'

    INSERT INTO DELETE_LOG (COLUMN_LIST)
    SELECT (
        SELECT *
        FROM DELETED
        FOR JSON PATH
    ) AS COLUMN_LIST;
END
```

```sql
DELETE FROM CRICKETER_DETAIL WHERE CDID = 35;
```

8%

No issues found

Messages

```
TRIGGER FIRED: TRG_CRICKETER_DETAIL_DELETE.

(1 row affected)
```

SQLQuery1....harsh (66))* | SQLQuery2.sq...\harsh (56))*

```sql
SELECT * FROM CRICKETER_DETAIL;

SELECT * FROM DELETE_LOG;
```

108 % 2 0

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS
1	1	Rohit	SHARMA	1987-04-30	Batter	121
2	2	VIRAT	KOHLI	1988-11-05	Batter	122
3	33	Yashasvi Jaiswal	Jaiswal	2001-12-28	Opening Batter	100
4	34	Rinku	Singh	1997-10-12	Middle-order Batter	NULL

	ID	COLUMN_LIST	CREATEDAT
1	1	[{"CDID":35,"FIRST_NAME":"Axar","LAST_NAME":"Pat...	2026-04-26 12:36:33.640

Part:19

```sql
--ANY COLUMN UPDATE

CREATE TABLE CRICKETER_DETAIL_ANY_COLUMN_UPDATE_LOG(
    ID INT IDENTITY(1,1) PRIMARY KEY,
    CDID INT,
    COLUMNNAME VARCHAR(100),
    OLDVALUE VARCHAR(MAX),
    NEWVALUE VARCHAR(MAX),
    CREATEDAT DATETIME DEFAULT GETDATE()
)
```

```sql
CREATE TRIGGER TRG_CRICKETER_DETAIL_ANY_COLUMN_UPDATE
ON CRICKETER_DETAIL
AFTER UPDATE
AS
BEGIN
    SET NOCOUNT ON;

    DECLARE @sql NVARCHAR(MAX);

    -- Move pseudo tables
    SELECT * INTO #I FROM INSERTED;
    SELECT * INTO #D FROM DELETED;
```

```sql
-- Build dynamic SQL safely
SELECT @sql = '
INSERT INTO CRICKETER_DETAIL_ANY_COLUMN_UPDATE_LOG (CDID, COLUMNNAME, OLDVALUE, NEWVALUE)
SELECT *
FROM (
    ' + STRING_AGG(
        '
        SELECT
            I.CDID,
            ''' + c.name + ''' AS COLUMNNAME,
            CONVERT(NVARCHAR(MAX), D.' + QUOTENAME(c.name) + ') AS OLDVALUE,
            CONVERT(NVARCHAR(MAX), I.' + QUOTENAME(c.name) + ') AS NEWVALUE
        FROM #I I
        JOIN #D D ON I.CDID = D.CDID
        ',
        ' UNION ALL '
    ) + '
) A
WHERE ISNULL(OLDVALUE, '''') <> ISNULL(NEWVALUE, '''');
'
```

```sql
    FROM sys.columns c
    WHERE c.object_id = OBJECT_ID('CRICKETER_DETAIL')
      AND c.name NOT IN ('CDID','CREATEDAT')
      AND c.is_computed = 0;

    -- DEBUG (optional)
    -- PRINT @sql;

    EXEC sp_executesql @sql;

    PRINT 'TRIGGER FIRED: TRG_CRICKETER_DETAIL_ANY_COLUMN_UPDATE.'

END;
```

SQLQuery1....harsh (66))* | SQLQuery2.sq...\harsh (56))*

```sql
SELECT * FROM CRICKETER_DETAIL;
```

108 % 1 0 Ln: 3, Ch: 1 TAB

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	Rohit	SHARMA	1987-04-30	Batter	121	2026-04-26 10:11:29.503	2026-04-26 12:24:37.610
2	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	2026-04-26 12:30:28.823
3	33	Yashasvi Jaiswal	Jaiswal	2001-12-28	Opening Batter	100	2026-04-26 12:10:49.323	2026-04-26 12:26:44.230
4	34	Rinku	Singh	1997-10-12	Middle-order Batter	NULL	2026-04-26 12:11:57.750	NULL

```sql
UPDATE CRICKETER_DETAIL
SET FIRST_NAME = 'Rohit S'
WHERE CDID = 1
```

```sql
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 1

SELECT * FROM CRICKETER_DETAIL_ANY_COLUMN_UPDATE_LOG WHERE CDID = 1
```

108 % 4 0

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	Rohit S	SHARMA	1987-04-30	Batter	121	2026-04-26 10:11:29.503	2026-04-26 12:41:50.133

	ID	CDID	COLUMNNAME	OLDVALUE	NEWVALUE	CREATEDAT
1	1	1	UPDATEDAT	Apr 26 2026 12:24PM	Apr 26 2026 12:41PM	2026-04-26 12:41:50.257
2	2	1	FIRST_NAME	Rohit	Rohit S	2026-04-26 12:41:50.267

```sql
UPDATE CRICKETER_DETAIL
SET FIRST_NAME = 'ROHIT'
WHERE CDID = 1
```

```sql
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 1

SELECT * FROM CRICKETER_DETAIL_ANY_COLUMN_UPDATE_LOG WHERE CDID = 1
```

108% 4 0

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	1	ROHIT	SHARMA	1987-04-30	Batter	121	2026-04-26 10:11:29.503	2026-04-26 12:43:18.993

	ID	CDID	COLUMNNAME	OLDVALUE	NEWVALUE	CREATEDAT
1	1	1	UPDATEDAT	Apr 26 2026 12:24PM	Apr 26 2026 12:41PM	2026-04-26 12:41:50.257
2	2	1	FIRST_NAME	Rohit	Rohit S	2026-04-26 12:41:50.267
3	3	1	UPDATEDAT	Apr 26 2026 12:41PM	Apr 26 2026 12:43PM	2026-04-26 12:43:19.073
4	4	1	FIRST_NAME	Rohit S	ROHIT	2026-04-26 12:43:19.080

```sql
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 2

SELECT * FROM CRICKETER_DETAIL_ANY_COLUMN_UPDATE_LOG WHERE CDID = 2
```

108 % 4 0

Results Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	2026-04-26 12:30:28.823

ID	CDID	COLUMNNAME	OLDVALUE	NEWVALUE	CREATEDAT

```sql
UPDATE CRICKETER_DETAIL
SET FIRST_NAME = 'Virat 18',HIGHEST_T20I_RUNS=122
WHERE CDID = 2
```

SQLQuery1....harsh (66))* | SQLQuery2.sq...\harsh (56))*

```sql
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 2

SELECT * FROM CRICKETER_DETAIL_ANY_COLUMN_UPDATE_LOG WHERE CDID = 2
```

108% 4 0

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	2	Virat 18	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	2026-04-26 12:44:47.60

	ID	CDID	COLUMNNAME	OLDVALUE	NEWVALUE	CREATEDAT
1	5	2	UPDATEDAT	Apr 26 2026 12:30PM	Apr 26 2026 12:44PM	2026-04-26 12:44:47.680
2	6	2	FIRST_NAME	VIRAT	Virat 18	2026-04-26 12:44:47.683

```sql
UPDATE CRICKETER_DETAIL
SET FIRST_NAME = NULL
WHERE CDID = 2
```

```sql
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 2

SELECT * FROM CRICKETER_DETAIL_ANY_COLUMN_UPDATE_LOG WHERE CDID = 2
```

108 % 4 0

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	2	NULL	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	2026-04-26 12:45:58.823

	ID	CDID	COLUMNNAME	OLDVALUE	NEWVALUE	CREATEDAT
1	5	2	UPDATEDAT	Apr 26 2026 12:30PM	Apr 26 2026 12:44PM	2026-04-26 12:44:47.680
2	6	2	FIRST_NAME	VIRAT	Virat 18	2026-04-26 12:44:47.683
3	7	2	UPDATEDAT	Apr 26 2026 12:44PM	Apr 26 2026 12:45PM	2026-04-26 12:45:58.897
4	8	2	FIRST_NAME	Virat 18	NULL	2026-04-26 12:45:58.903

```sql
UPDATE CRICKETER_DETAIL
SET FIRST_NAME = 'VIRAT'
WHERE CDID = 2
```

```sql
SELECT * FROM CRICKETER_DETAIL WHERE CDID = 2

SELECT * FROM CRICKETER_DETAIL_ANY_COLUMN_UPDATE_LOG WHERE CDID = 2
```

98% 4 0

Results | Messages

	CDID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS	CREATEDAT	UPDATEDAT
1	2	VIRAT	KOHLI	1988-11-05	Batter	122	2026-04-26 10:11:29.513	2026-04-26 12:46:35.667

	ID	CDID	COLUMNNAME	OLDVALUE	NEWVALUE	CREATEDAT
1	5	2	UPDATEDAT	Apr 26 2026 12:30PM	Apr 26 2026 12:44PM	2026-04-26 12:44:47.680
2	6	2	FIRST_NAME	VIRAT	Virat 18	2026-04-26 12:44:47.683
3	7	2	UPDATEDAT	Apr 26 2026 12:44PM	Apr 26 2026 12:45PM	2026-04-26 12:45:58.897
4	8	2	FIRST_NAME	Virat 18	NULL	2026-04-26 12:45:58.903
5	9	2	UPDATEDAT	Apr 26 2026 12:45PM	Apr 26 2026 12:46PM	2026-04-26 12:46:35.733
6	10	2	FIRST_NAME	NULL	VIRAT	2026-04-26 12:46:35.740